

Assignment # 2 :- Convert the regular expressions into CFG :-

1. regular expression

$$(b^* (aaa)^+)^* b^*$$

$$S \rightarrow bS$$

$$S \rightarrow aaaS$$

$$S \rightarrow \epsilon$$

2. regular expression

$$a^* b a^* b a^* b a^*$$

$$S \rightarrow AbAbAbA$$

$$A \rightarrow aA \mid \epsilon$$

3. regular expression

$$(a+b)^* aa + (a+b)^* bb$$

$$S \rightarrow Aaa \mid Abb$$

$$A \rightarrow aA \mid bA \mid \epsilon$$

4. regular expression

$$(a+b)^* aa (b(a+b)^*) + (a+b)^* bb (a(a+b)^*)$$

$$S \rightarrow XaabX \mid XbbaX$$

$$X \rightarrow aX \mid bX \mid \epsilon$$

5. regular expression

$$(a + ba + bba)^*$$

$$S \rightarrow aS \mid baS \mid bbaS \mid \epsilon$$

6. regular expression

$$(a + b)(aa + bb + ab + ba)^*$$

$$S \rightarrow aX \mid bX$$

$$X \rightarrow aaX \mid bbX \mid abX \mid baX \mid \epsilon$$

7. regular expression

$$(a + b)^*(aa + bb)$$

$$S \rightarrow Xaa \mid Xbb$$

$$X \rightarrow aX \mid bX \mid \epsilon$$

8. regular expression

$$(a + b)^*(aa + bb) + (aa + bb)(a + b)^*$$

$$S \rightarrow Xaa \mid Xbb \mid aaX \mid bbX$$

$$X \rightarrow aX \mid bX \mid \epsilon$$

9. regular expression

$$(a + b)^*a(a + b) + (ba + ab)^*(a + b)b$$

$$S \rightarrow Xaa \mid Xab \mid Zab \mid Zbb$$

$$X \rightarrow aX \mid bX \mid \epsilon$$

$$Z \rightarrow baZ \mid abZ \mid \epsilon$$

10. regular expression

$$(a + b)^*a(ab + ba) + (ba + ab)^*(a + b)b$$

$S \rightarrow Xaab \mid Xaba \mid Yab \mid Ybb$

$X \rightarrow aX \mid bX \mid \epsilon$

$Y \rightarrow baY \mid abY \mid ba \mid ab$

Convert CFL to CFG

1. cfl

$L = \{w \mid w \in \{0,1\}^* \text{ and } w \text{ start, and end with the same letter}\}$

$S \rightarrow 0A0 \mid 1A1 \mid 0 \mid 1$

$A \rightarrow 0A \mid 1A \mid \epsilon$

2. cfl

$L = \{w \mid w \in \{0,1\}^* \text{ and } w \text{ is odd}\}$

$S \rightarrow A1$

$A \rightarrow 0A \mid 1A \mid \epsilon$

3. cfl

$L = \{a^n b a^n \mid n > 1\}$

$S \rightarrow aSa \mid aabaa$

4. cfl

$L = \{a^n b c^{2n} \mid n > 1\}$

$S \rightarrow aSc \mid aabcccc$

5. cfl

$$L = \{ a^n b a^n \mid n > 1 \}$$

$$S \rightarrow aSa \mid aabaa$$

6. cfl

$$L = \{ w \mid w = w^R \}$$

$$S \rightarrow aSa \mid bSb \mid a \mid b \mid \epsilon$$

Simplify the CFG

case 1 :-

$$S \rightarrow ABC$$

$$A \rightarrow aA \mid \epsilon$$

$$B \rightarrow bB \mid \epsilon$$

$$C \rightarrow c$$

$$S \rightarrow ABC \mid BC \mid AC \mid c$$

$$A \rightarrow aA \mid a$$

$$B \rightarrow bB \mid b$$

$$C \rightarrow c$$

case 2 :-

$$S \rightarrow XY$$

$$X \rightarrow a$$

$$Y \rightarrow Z \mid b$$

$$Z \rightarrow M$$

$$M \rightarrow N$$

$$N \rightarrow a$$

$$S \rightarrow XY$$

$$X \rightarrow a$$

$$Y \rightarrow a \mid b$$

case 3 :-

$$S \rightarrow ASA \mid b$$

$$A \rightarrow B \mid AB \mid \epsilon$$

$$B \rightarrow AC \mid b$$

$$D \rightarrow d$$

$$S \rightarrow ASA \mid AS \mid SA \mid b$$

$$A \rightarrow b \mid AB$$

$$B \rightarrow b$$

case 4 :-

$$S \rightarrow abS \mid abA$$

$$A \rightarrow cd$$

$$c \rightarrow dc$$

$S \rightarrow abS \mid abA$

$A \rightarrow cd$

case 5:-

$S \rightarrow aA$

$A \rightarrow a$

$A \rightarrow B$

$B \rightarrow a$

$B \rightarrow bb$

$S \rightarrow aA$

$A \rightarrow a \mid bb$